

127. What is the value of $\frac{1}{\sqrt{5} + \sqrt{3}} + \frac{2}{3 + \sqrt{5}} - \frac{3}{3 + \sqrt{3}}$?

(I.I.F.T., 2005)

- (a) $-\frac{1}{2}$ (b) 0
(c) $\frac{1}{2}$ (d) 1

128. If $\sqrt{2} = 1.4142$, the value of $\frac{7}{(3 + \sqrt{2})}$ is (R.R.B., 2005)

- (a) 1.5858 (b) 3.4852
(c) 3.5858 (d) 4.4142

129. $\left[\frac{3\sqrt{2}}{\sqrt{6} - \sqrt{3}} - \frac{4\sqrt{3}}{\sqrt{6} - \sqrt{2}} - \frac{6}{\sqrt{8} - \sqrt{12}} \right] = ?$

(Teachers' Exam, 2010)

- (a) $\sqrt{3} - \sqrt{2}$ (b) $\sqrt{3} + \sqrt{2}$
(c) $5\sqrt{3}$ (d) 1

130. $\frac{\sqrt{7} + \sqrt{5}}{\sqrt{7} - \sqrt{5}}$ is equal to

- (a) 1 (b) 2
(c) $6 - \sqrt{35}$ (d) $6 + \sqrt{35}$

131. If $\frac{5 + 2\sqrt{3}}{7 + 4\sqrt{3}} = a + b\sqrt{3}$, then (R.R.B., 2010)

- (a) $a = -11, b = -6$ (b) $a = -11, b = 6$
(c) $a = 11, b = -6$ (d) $a = 6, b = 11$

132. If $\sqrt{2} = 1.414$, the square root of $\frac{\sqrt{2} - 1}{\sqrt{2} + 1}$ is nearest

to

- (a) 0.172 (b) 0.414
(c) 0.586 (d) 1.414

133. Given that $\sqrt{3} = 1.732$, the value of

$\frac{3 + \sqrt{6}}{5\sqrt{3} - 2\sqrt{12} - \sqrt{32} + \sqrt{50}}$ is (S.S.C., 2007)

- (a) 1.414 (b) 1.732
(c) 2.551 (d) 4.899

134. $\left(\frac{2 + \sqrt{3}}{2 - \sqrt{3}} + \frac{2 - \sqrt{3}}{2 + \sqrt{3}} + \frac{\sqrt{3} - 1}{\sqrt{3} + 1} \right)$ simplifies to (S.S.C., 2008)

- (a) $16 - \sqrt{3}$ (b) $4 - \sqrt{3}$
(c) $2 - \sqrt{3}$ (d) $2 + \sqrt{3}$

135. If $x = \frac{\sqrt{5} + \sqrt{3}}{\sqrt{5} - \sqrt{3}}$ and $y = \frac{\sqrt{5} - \sqrt{3}}{\sqrt{5} + \sqrt{3}}$, then $(x + y)$

equals

(S.S.C., 2005)

- (a) $2(\sqrt{5} + \sqrt{3})$ (b) $2\sqrt{15}$
(c) 8 (d) 16

136. $\frac{\sqrt{2}(2 + \sqrt{3})}{\sqrt{3}(\sqrt{3} + 1)} \times \frac{\sqrt{2}(2 - \sqrt{3})}{\sqrt{3}(\sqrt{3} - 1)}$ is equal to (C.P.O., 2006)

- (a) $3\sqrt{2}$ (b) $\frac{\sqrt{2}}{3}$
(c) $\frac{2}{3}$ (d) $\frac{1}{3}$

137. $\frac{12}{3 + \sqrt{5} + 2\sqrt{2}}$ is equal to (S.S.C., 2007)

- (a) $1 - \sqrt{5} + \sqrt{2} + \sqrt{10}$
(b) $1 + \sqrt{5} + \sqrt{2} - \sqrt{10}$
(c) $1 + \sqrt{5} - \sqrt{2} + \sqrt{10}$
(d) $1 - \sqrt{5} - \sqrt{2} + \sqrt{10}$

138. $\left[\frac{1}{\sqrt{2} + \sqrt{3} - \sqrt{5}} + \frac{1}{\sqrt{2} - \sqrt{3} - \sqrt{5}} \right]$ in simplified form

equals

(S.S.C., 2005)

- (a) 0 (b) $\frac{1}{\sqrt{2}}$
(c) 1 (d) $\sqrt{2}$

139. If $x = (7 - 4\sqrt{3})$, then the value of $\left(x + \frac{1}{x} \right)$ is

- (a) $3\sqrt{3}$ (b) $8\sqrt{3}$
(c) 14 (d) $14 + 8\sqrt{3}$

140. If $x = 3 + \sqrt{8}$, then $x^2 + \frac{1}{x^2}$ is equal to (S.S.C., 2007)

- (a) 30 (b) 34
(c) 36 (d) 38

141. If $a = \frac{\sqrt{3} + \sqrt{2}}{\sqrt{3} - \sqrt{2}}$, $b = \frac{\sqrt{3} - \sqrt{2}}{\sqrt{3} + \sqrt{2}}$, then the value of

$a^2 + b^2$ would be

(M.B.A., 2008)

- (a) 10 (b) 98
(c) 99 (d) 100

142. If $a = \frac{\sqrt{5} + 1}{\sqrt{5} - 1}$ and $b = \frac{\sqrt{5} - 1}{\sqrt{5} + 1}$, the value

of $\left(\frac{a^2 + ab + b^2}{a^2 - ab + b^2} \right)$ is

- (a) $\frac{3}{4}$ (b) $\frac{4}{3}$
(c) $\frac{3}{5}$ (d) $\frac{5}{3}$